

Normal Laboratory Reference Intervals

Included here are some of the most common tests and their reference intervals (formally referred to as *normal values*). Laboratory values may vary depending on different techniques or different laboratories.

Laboratory Test	Normal Reference Interval
Arterial Blood Gases	
pH	7.35-7.45
PCO ₂	35-45 mm Hg
HCO ₃ ⁻	22-27 mm Hg
PO ₂	80%-100%
SaO ₂	96%-100%
Renal Function Studies	
Blood urea nitrogen (BUN)	8-25 mg/dL
Creatinine	0.6-1.3 mg/dL
Gastrointestinal Studies	
Cholesterol	Total: 140-199 mg/dL
	High-density lipoprotein (HDL): 30-70 mg/dL
	Low-density lipoprotein (LDL): less than 130 mg/dL
Triglycerides	Lower than 200 mg/dL
Protein	6-8 g/dL
Albumin	3.4-5 g/dL
Amylase	25-151 units/L
Lipase	10-140 units/L
Ammonia	10-80 mcg/dL
Alanine aminotransferase (ALT)	10-40 units/L
Aspartate aminotransferase (AST)	10-30 units/L
Bilirubin	• Bilirubin, direct (conjugated): 0-0.3 mg/dL • Bilirubin, indirect (unconjugated): 0.1-1 mg/dL • Bilirubin, total: lower than 1.5 mg/dL

Laboratory Test	Normal Reference Interval
Glucose Studies	
Glucose	70-110 mg/dL
Hemoglobin A _{1c}	4%-5.9%: nondiabetic 7% or lower: good diabetic control 7% to 8%: fail diabetic control Higher than 8%: poor diabetic control
Erythrocyte Studies	
Hemoglobin	Female: 12-15 g/dL
	Male: 14-16.5 g/dL
Hematocrit	Female: 35%-47%
	Male: 42%-52%
Coagulation Studies	
Platelets	150,000-400,000 cells/mm ³
Activated partial thromboplastin time (aPTT)	20-36 sec, depending on testing method
Prothrombin time (PT)	9.5-11.8 sec
International normalized ratio (INR)	• 2 to 3 for standard warfarin therapy • 3 to 4.5 for high-dose warfarin therapy
Electrolyte Values	
Potassium	3.5-5.0 mEq/L
Sodium	135-145 mEq/L
Chloride	98-107 mEq/L
Bicarbonate (venous)	22-29 mEq/L

Laboratory Test	Normal Reference Interval
White Blood Cell and Differential Count	
White blood cells	4500-11,000 cells/mm ³
Neutrophils	1800-7800 cells/mm ³
Bands	0-700 cells/mm ³
Eosinophils	0-450 cells/mm ³
Basophils	0-200 cells/mm ³
Lymphocytes	1000-4800 cells/mm ³
Monocytes	0-800 cells/mm ³
Cardiac Markers	
Creatine kinase (CK)	CK-MB: 0%-5% of total CK value
	CK-MM: 95%-100% of total CK value
	CK-BB: 0% of CK value
Troponin I	Lower than 0.6 ng/mL
Troponin T	Lower than 0.1-0.2 ng/mL
Myoglobin	Lower than 90 mcg/L
Natriuretic peptides	Atrial natriuretic peptides (ANP): 22-27 pg/mL Brain natriuretic peptides (BNP): less than 100 pg/mL C-type natriuretic peptides (CNP): not determined
Elements	
Magnesium	1.6-2.6 mg/dL
Phosphorus	2.7-4.5 mg/dL
Calcium	8.6-10 mg/dL